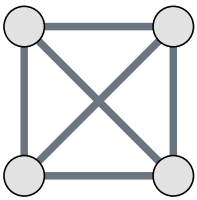
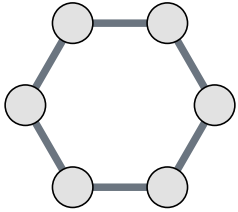
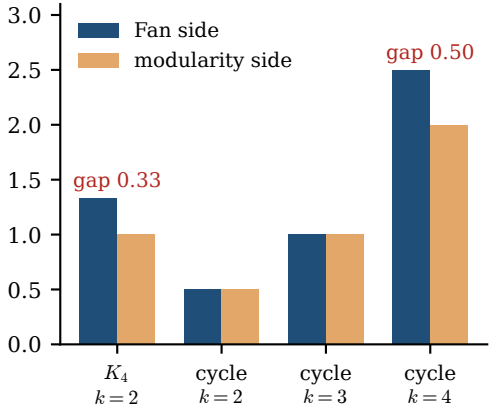
(a) K_4, weights $1/3$
every degree 1



(b) 6-cycle, weights $1/2$
every degree 1



(c) the two sides of claim (b)